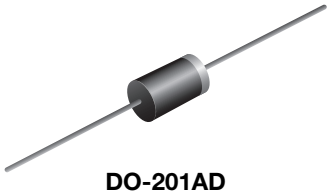
# Soft Recovery Ultrafast Plastic Rectifier



DO-201AD

## FEATURES

- Glass passivated pellet chip junction
- Ultrafast reverse recovery time
- Low forward voltage drop
- Low switching losses, high efficiency
- High forward surge capability
- Solder dip 275 °C max. 10 s, per JESD 22-B106
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT

## TYPICAL APPLICATIONS

For use in high frequency rectification and freewheeling application in switching mode converters and inverters for consumer, computer and telecommunication.

## MECHANICAL DATA

**Case:** DO-201AD
Molding compound meets UL 94 V-0 flammability rating
Base P/N-E3 - RoHS-compliant, commercial grade

**Terminals:** matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
E3 suffix meets JESD 201 class 1A whisker test

**Polarity:** color band denotes cathode end

| PRIMARY CHARACTERISTICS | |
|---|---|
| $I_{F(AV)}$ | 3.0 A |
| $V_{RRM}$ | 50 V, 100 V, 200 V, 300 V, 400 V, 500 V, 600 V, 800 V, 1000 V |
| $I_{FSM}$ | 150 A |
| $t_{rr}$ | 50 ns, 75 ns |
| $V_F$ | 1.0 V, 1.7 V |
| $T_J$ max. | 150 °C |
| Package | DO-201AD |
| Circuit configuration | Single |

**MAXIMUM RATINGS** ($T_A$ = 25 °C unless otherwise noted)

| PARAMETER | SYMBOL | UF5400 | UF5401 | UF5402 | UF5403 | UF5404 | UF5405 | UF5406 | UF5407 | UF5408 | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Maximum repetitive peak reverse voltage | $V_{RRM}$ | 50 | 100 | 200 | 300 | 400 | 500 | 600 | 800 | 1000 | V |
| Maximum RMS voltage | $V_{RMS}$ | 35 | 70 | 140 | 210 | 280 | 350 | 420 | 560 | 700 | V |
| Maximum DC blocking voltage | $V_{DC}$ | 50 | 100 | 200 | 300 | 400 | 500 | 600 | 800 | 1000 | V |
| Maximum average forward rectified current, 0.375" (9.5 mm) lead length at $T_A$= 55 °C | $I_{F(AV)}$ | 3.0 | | | | | | | | | A |
| Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load | $I_{FSM}$ | 150 | | | | | | | | | A |
| Operating junction and storage temperature range | $T_J$, $T_{STG}$ | -55 to +150 | | | | | | | | | °C |


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| **ELECTRICAL CHARACTERISTICS** ($T_A$ = 25 °C unless otherwise noted) | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **PARAMETER** | **TEST CONDITIONS** | | **SYMBOL** | **UF5400** | **UF5401** | **UF5402** | **UF5403** | **UF5404** | **UF5405** | **UF5406** | **UF5407** | **UF5408** | **UNIT** |
| Maximum instantaneous forward voltage | 3.0 A | | $V_F$ (1) | 1.0 | | | | | 1.7 | | | | V |
| Maximum DC reverse current at rated DC blocking voltage | | $T_A$ = 25 °C | $I_R$ | 10 | | | | | | | | | μA |
| | | $T_A$ = 100 °C | | 75 | | | | | 200 | | | | |
| Maximum reverse recovery time | $I_F$ = 0.5 A, $I_R$ = 1.0 A, $I_{rr}$ = 0.25 A | $T_J$ = 25 °C | $t_{rr}$ | 50 | | | | | 75 | | | | ns |
| Typical junction capacitance | 4.0 V, 1 MHz | | $C_J$ | 45 | | | | | 36 | | | | pF |

**Note**

(1) Pulse test: 300 μs pulse width, 1 % duty cycle

| **THERMAL CHARACTERISTICS** ($T_A$ = 25 °C unless otherwise noted) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **PARAMETER** | **SYMBOL** | **UF5400** | **UF5401** | **UF5402** | **UF5403** | **UF5404** | **UF5405** | **UF5406** | **UF5407** | **UF5408** | **UNIT** |
| Typical thermal resistance | $R\theta_{JA}$ (1) | 20 | | | | | | | | | °C/W |
| | $R\theta_{JL}$ (1) | 8.5 | | | | | | | | | |

**Note**

(1) Thermal resistance from junction to lead and from junction to ambient with 0.375" (9.5 mm) lead length, both leads attached to heatsink

| **ORDERING INFORMATION** (Example) | | | | |
|---|---|---|---|---|
| **PREFERRED P/N** | **UNIT WEIGHT (g)** | **PREFERRED PACKAGE CODE** | **BASE QUANTITY** | **DELIVERY MODE** |
| UF5406-E3/54 | 1.172 | 54 | 1400 | 13" diameter paper tape and reel |
| UF5406-E3/73 | 1.172 | 73 | 1000 | Ammo pack packaging |

## RATINGS AND CHARACTERISTICS CURVES ($T_A$ = 25 °C unless otherwise noted)

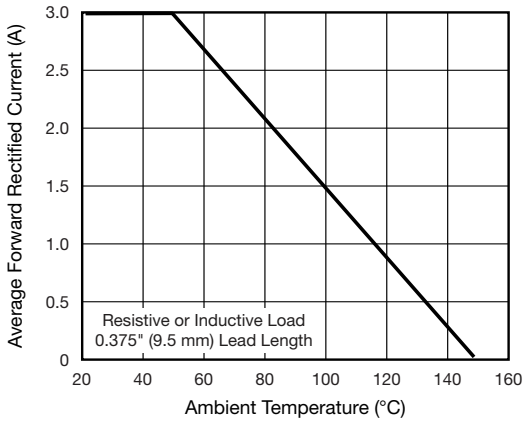



Fig. 1 - Maximum Forward Current Derating Curve

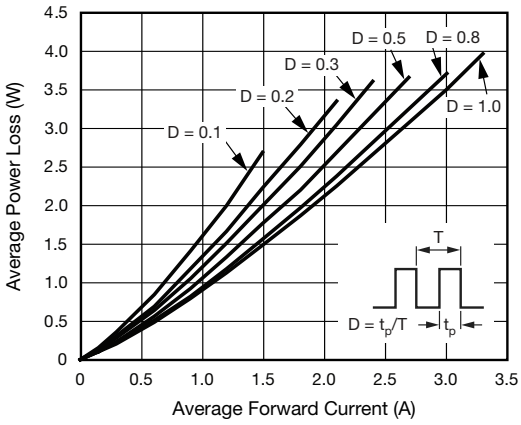



Fig. 2 - Forward Power Loss Characteristics


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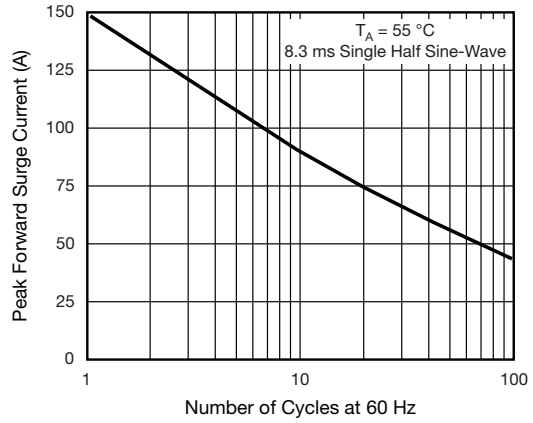



Fig. 3 - Maximum Non-Repetitive Peak Forward Surge Current

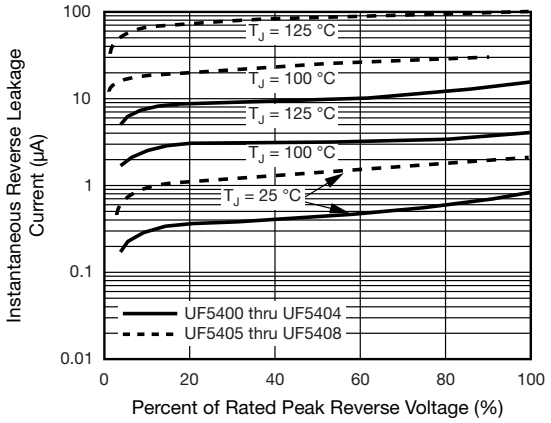



Fig. 5 - Typical Reverse Leakage Characteristics

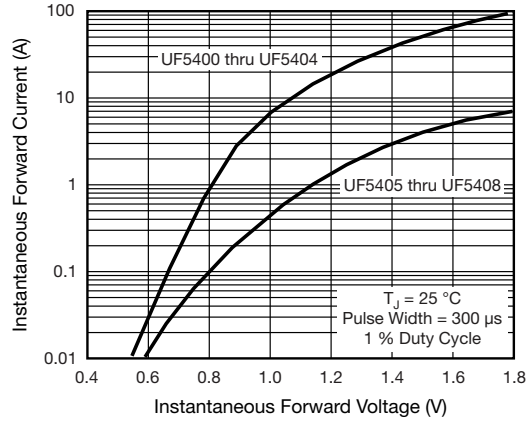



Fig. 4 - Typical Instantaneous Forward Characteristics

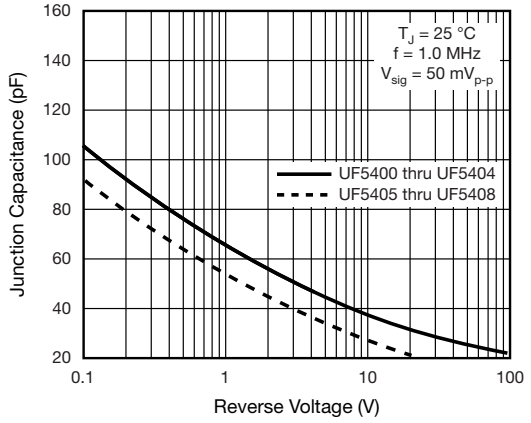
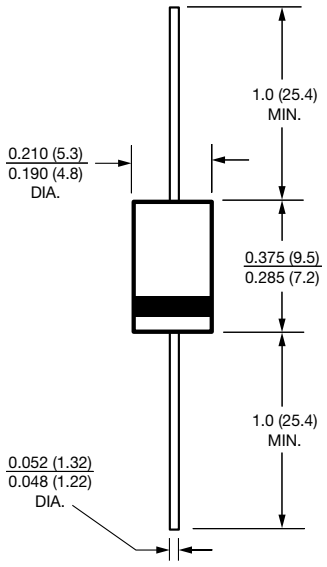



Fig. 6 - Typical Junction Capacitance

## **PACKAGE OUTLINE DIMENSIONS** in inches (millimeters)

**DO-201AD**





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